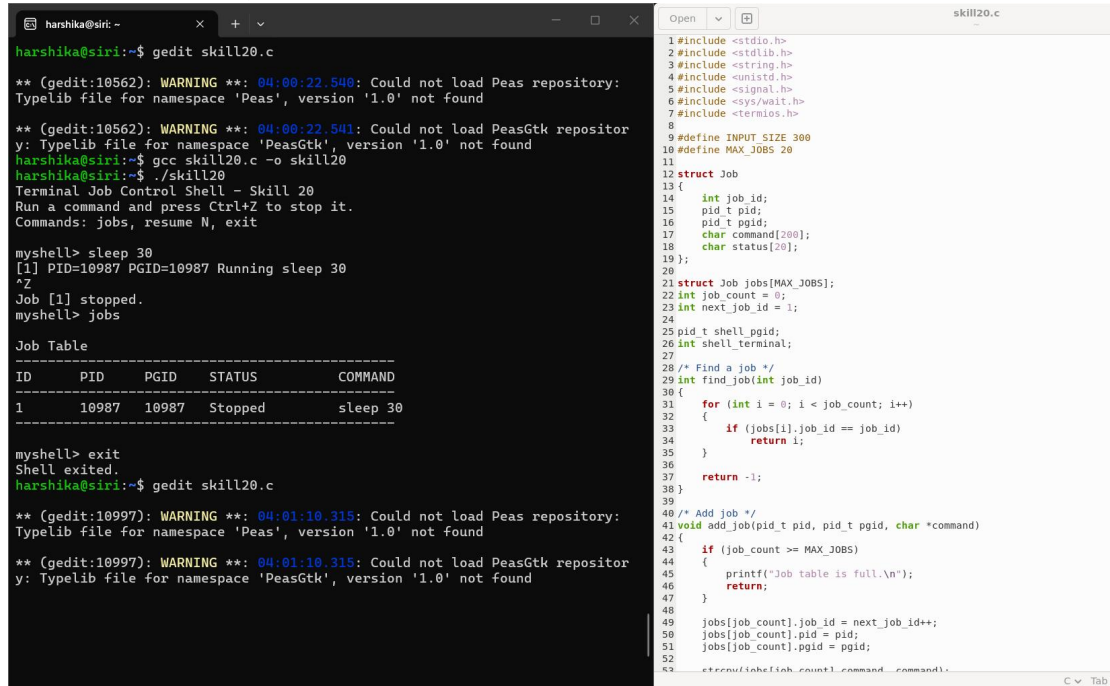


Skill - 20

1. Process Groups and Terminal Controls



The screenshot shows a terminal window on the left and a code editor on the right. The terminal window displays the execution of a C program named `skill20.c`. The user runs `gcc skill20.c -o skill20` and then `./skill20`. The program outputs "Terminal Job Control Shell - Skill 20" and "Run a command and press Ctrl+Z to stop it. Commands: jobs, resume N, exit". The user enters `sleep 30`, and the program outputs "[1] PID=10987 PGID=10987 Running sleep 30". The user presses Ctrl+Z, and the program outputs "Job [1] stopped." and "myshell> jobs". The program then displays a "Job Table" with the following content:

ID	PID	PGID	STATUS	COMMAND
1	10987	10987	Stopped	sleep 30

The user enters `exit`, and the program outputs "Shell exited." and "myshell> jobs". The code editor on the right shows the source code of `skill20.c`, which is a C program for process control. It includes headers for `stdio.h`, `stdlib.h`, `string.h`, `unistd.h`, `signal.h`, `sys/wait.h`, and `termios.h`. It defines `INPUT_SIZE` as 300 and `MAX_JOBS` as 20. It uses a `struct Job` to store job information and a `jobs` array to store job data. The program implements a simple job control shell with commands for running, stopping, and listing jobs.